

Run three machine cycles

Name _____

Your job: run three instructions. Each one goes through the same four stages — **FETCH** → **DECODE** → **EXECUTE** → **WRITE**. The instructions are shortened codes, so you have to look each one up in the table to find out what it means.

MEMORY	STARTING STATE	DECODE LOOKUP TABLE															
0: LDI 5	PC = 0	<table border="1"><thead><tr><th>Code</th><th>What it means</th><th>Destination</th></tr></thead><tbody><tr><td>LDI n</td><td>Load the number n into register A</td><td>A</td></tr><tr><td>ADI n</td><td>Add the number n to what is in A</td><td>A</td></tr><tr><td>SBI n</td><td>Subtract the number n from what is in A</td><td>A</td></tr><tr><td>STA n</td><td>Copy what is in A into memory address n</td><td>M[n]</td></tr></tbody></table>	Code	What it means	Destination	LDI n	Load the number n into register A	A	ADI n	Add the number n to what is in A	A	SBI n	Subtract the number n from what is in A	A	STA n	Copy what is in A into memory address n	M[n]
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1: ADI 1	IR = empty																
2: STA 20	A = unset																
20: (data – empty)	M[20] = empty																

The trace — one row per instruction, one column per stage

Instruction	1. FETCH Copy the instruction at the PC into the IR, then move the PC on.	2. DECODE Look the code up. What does it mean, and where does the result go?	3. EXECUTE Work out the result. Nothing is stored yet.	4. WRITE Put the result in the destination you decoded.
Cycle 1 PC starts at 0	IR holds _____ PC is now _____	Means _____ Destination _____	Result _____	A = _____
Cycle 2 use the PC you wrote	IR holds _____ PC is now _____	Means _____ Destination _____	Calculation _____ Result _____	A = _____
Cycle 3 use the PC you wrote	IR holds _____ PC is now _____	Means _____ Destination _____	Value to be copied _____ _____	Write it here: _____

Stuck on a box? Every box has its own hint on the Canvas page for this activity.

Final check

Ending state: PC = _____ A = _____ M[20] = _____

You ran **three** instructions. How many stages did each one pass through? _____

In cycle 3 the result did not go into register A. Which stage told you that?

Why could the machine not skip the decode stage on this worksheet?

Model boundary: a real machine stores instructions as numbers, not as short codes like **LDI**, and its lookup table is built into the hardware rather than printed on a page. We are hiding that encoding so the repeating cycle stays visible.